

Packager Guidelines

Section 7: Compressor Gas Piping

7-0. GENERAL

We cannot overemphasize the necessity for thoroughly cleaning the inside of all piping before running the unit, whether during shop test or after installation in the field. Most packagers use welded carbon steel piping, which often contains mill scale and other debris. It is particularly important that the cylinder intake piping and start-up bypass lines be thoroughly clean. If metal shavings, pipe scale, rust, welding beads, dirt and all other foreign materials are not carefully and thoroughly removed, the gas flow will loosen this material and carry it into the compressor cylinders causing damage and/or excessive wear. Piping should be fabricated using good shop practice and inspected and cleaned in accordance with industry standards.

Piping should be installed so that there is no stress on attachment points. Once in place, all flange fasteners should be removed to confirm that piping fits without being forced into place. It is important that all piping be properly braced to avoid hanging excessive weight from cylinder connections.

All flanged cylinder connections are closed for shipment. Do not remove these closures until the mating piping is ready for assembly. If plugs or covers on any of the compressor connections, or other openings, have become loosened or lost in transit, the inside of the component must be thoroughly wiped out to remove any dirt.

7-1 Cylinder Flange Inlet and Discharge Studs

Cylinders are supplied with Inlet and Discharge (I&D) studs and nuts. Cylinder I&D thread projection is dependent on the style of the cylinder and associated flange size/rating. Unless specified otherwise by the order, the following methods are used to determine the minimum full thread projections.

1. Cast Cylinders with standard ANSI FF Flange and size ratings
 - a. I&D thread projections are based on the sum of the industry standard ANSI RF flange thickness, the raised face, the supplied nut height, an additional 5/8" (15.875 mm) (to account for 2 gaskets at 1/8" (3.175 mm) and 1 orifice plate at 3/8" (9.525 mm), and an additional 1/4" (6.4 mm). This should leave at least one exposed thread on each I&D stud after the nut is torqued.
2. Forged Steel Cylinders with RTJ (Ring Type Joint) flanges
 - a. I&D thread projections are based on the sum of the industry standard ANSI RTJ flange thickness, the corresponding oval style ring height, the supplied nut height, an additional 3/8" (9.525 mm) to account for the plate thickness in a 1-piece ring and orifice plate assembly, and an additional 1/4" (6.4 mm). This should leave at least one exposed thread on each I&D stud after the nut is torqued.
3. Forged Steel Cylinders with 3.00" (76.2 mm)-6000# (26.69 kN) Special Lens Ring Joint
 - a. I&D Special RF Weld Neck Flanges and special lens ring gaskets are supplied. The thread projection is based on the sum of the flange thickness, 1/4" (6.4 mm) raised face, gasket gap, supplied nut height, and an additional 1/4" (6.4 mm). This should leave at least one exposed thread on each I&D stud after the nut is torqued. If the application requires an orifice based on the pulsation analysis, it can be incorporated into the lens ring gasket machining upon request.
4. Application on Specific Cylinders with Special Flanges
 - a. Occasionally, non-standard, application specific cylinders and/or flanges are designed and the I&D thread projection will be communicated via Project Engineering.
 - b. 26-28" (660.4-711.2 mm) HOS/HOSS Cylinders have special rectangular I&D flanges due to the geometry. The I&D thread projection will be communicated via Project Engineering.

7-2. Cylinder Inlet and Discharge Gasket Recommendations

Siemens Energy offers the following recommendations for gaskets on cast iron cylinders with AISI FF Inlet and Discharge flange connections:

1. For Class 600 and below, use Flexitallic® (or equivalent) Style CG gaskets
2. For Class 900 and above, use Flexitallic® (or equivalent) Style CGI gaskets

7-3. Gas Inlet Piping

The gas must be clean and dry before it enters the compressor. Scrubbers or separators of suitable size are often used for this purpose. Carefully arrange the suction or intake piping to prevent loops or pockets in the line. This will avoid the collection of condensate or distillate, which could be drawn into the compressor cylinder and cause serious damage to the unit. Use of suction screens during start-up to filter out pipeline debris and dirt is essential. In some applications, more elaborate filtration systems, even coalescers, may be necessary.

On multi-stage installations precautions should be taken to prevent condensate carry over to the subsequent cylinders. Scrubbers or separators should be provided in the suction piping of second and subsequent stage cylinders.

7-4. Discharge Piping

The discharge line between the compressor and receiver or aftercooler should drain away from the cylinder, and should be as short as possible, avoiding short bends and fittings.

A bypass with proper valving should be located between the compressor discharge line and intake line to aid in starting and stopping the unit. With such a bypass installed around the cylinder and the valves properly manipulated, the bypass opening will reduce load on the unit and allow starting or stopping the unit against minimal load. If the unit is using a bypass valve for capacity control, the suction scrubbers or separators must be suitable to clean the gas along with removing the lubricator oil that is entrained in the gas after the last stage of compression.

Relief valves should be installed in the interstage and discharge lines. The setting should be less than or equal to the lowest maximum allowable working pressure of any pressure-containing component in the system being protected.

Install a drain valve at the lowest point in the discharge line to permit periodic draining of liquids that may accumulate.

7-5. Temporary Suction Screens**WARNING**

The packager should ensure that the operator has installed a temporary start-up screen. This is required to catch any dirt, chips, or other foreign material that may have been left in the pipe. The pipe must be cleaned first, however, as dependence on the temporary line filter alone is not sufficient.

If there is any possibility that the gas being compressed contains dust, sand, or other abrasive particles, a gas scrubber must be installed permanently and then regularly serviced to keep it clean.

Even though the previous cleaning procedure has been carefully followed on the compressor piping, a temporary screen should be installed in the intake line to remove particles 0.009 inch (0.23 mm) (230 microns) in diameter or larger. Provision must also be made in the piping to check the pressure drop across the screen and to permit removal of the screen for cleaning. If the pressure drop across the screen exceeds 5% of the upstream line pressure, remove the screen, clean it thoroughly, and reinstall it. The screen should be removed and left out only when the inlet line is completely free of welding beads, pipe scale, and other extraneous matter. These screens are not intended for permanent installation. We recommend that the screen be installed in a horizontal run of piping to facilitate cleaning out any material that collects on the upstream side.

7-6. Safety Valves

The compressor must be fitted with safety valves to limit suction pressure, interstage pressure and final discharge pressure to a safe maximum. For this application, the term "safety valve" refers to an automatic, pressure-relieving device that is actuated by the static pressure upstream of the valve and characterized by full opening pop action; these are also often referred to as "pop safety valves."

CAUTION

Always install a safety valve capable of passing the full-load capacity of the compressor cylinder, or cylinders, between the cylinder and the first line valve. UNDER NO CIRCUMSTANCES is a shutoff valve to be installed in any discharge line before the safety valve.

When a bypass is furnished, a suction pressure safety valve should be installed immediately downstream of the bypass valve, on the inlet scrubber, or suction piping. This valve must be between the line suction valve and the unit.

For recommended safety settings, see chart below:

Relief Valve Settings

Rated Discharge Gauge Pressure (Each Stage)	Minimum Relief Valve Set Pressure Margin Above Rated Discharge Gauge Pressure
PSIG (kg/cm ²)	PSIG (kg/cm ²)
≤150 (10.5)	15 (1.1)
>150 to 2500 (10.5 to 175.8)	10%
>2500 to 3500 (175.8 to 246.1)	8%
>3500 to 5000 (246.1 to 351.3)	6%
>5000 (351.5)	See Footnote A
a) For rated discharge gauge pressures above 5000 psig (351.5 kg/cm ²), the relief valve shall be agreed on by the purchaser and the vendor.	

NOTE

The pressure at which any safety valve is set to operate must include the effects of static head and constant back pressure. Safety valves are selected for the required pressure setting and also for expected gas characteristics and temperature.

NOTE

Rupture discs may be used in place of, or supplementary to, safety valves on certain applications. These are special cases where the proper action of a safety valve might be questionable because of corrosion or deposits caused by the process, where a loss of valuable material by leakage should be avoided, or where contamination of the atmosphere by leakage of noxious gases must be avoided.

Safety valves must be constructed, located and installed so that they are readily accessible for inspection and repair and so that they cannot be readily rendered inoperative. They must also meet any Federal and state or local regulations.

Discharge lines from safety valves must be designed to facilitate drainage, and such lines must lead to a safe place of discharge. The size of the discharge line must be such that any back pressure that may exist or develop will not reduce the relieving capacity of the safety valve below that required to properly protect the compressor.

Further information pertaining to the selection, pressure settings, installation and piping of safety valve and other relief devices will be found in one of the pressure vessel codes.

7-7. Intercooler, Aftercooler, Scrubber and Frame

7-7.1. Intercooler

Intercooling will condense moisture or vapor out of the gas. This moisture must be drained from the cooler to prevent its passing into the high-pressure cylinder where it would interfere with lubrication. It is important that a schedule is established whereby the intercooler is regularly drained as determined by the local conditions, or proper scrubbing equipment is installed.

Intercoolers used on gas field separable applications are generally the fin-fan type (gas to air), using a heat exchanger in conjunction with a fan drive and, possibly, louvers controlled by a shutterstat. See the applicable manufacturer's literature for a description and service procedures.

7-7.2. Aftercooler

For some compressor applications, it is desirable to have an aftercooler to cool the discharge gases and to condense any moisture carried by the gas. This prevents the moisture from condensing later in the lines where it might freeze, cause water hammer, or give other trouble.

7-7.3. Scrubber

Skid-mounted scrubbers are not normally adequate for proper cleanup of extremely wet or contaminated field gas. Siemens Energy recommends all scrubbers have automatic dumps as well as manual dump connections. Siemens Energy also recommends all scrubbers have sight glasses. Gas having these properties almost always will cause maintenance problems with the compressor or driver. It is recommended that an off-skid mounted scrubber be installed if such conditions exist.

7-7.4. Frame

If the compressor is located inside a building and pumping explosive or toxic gas, the frame should be vented to the outside of the building.